

Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. A network administrator wants to block incoming traffic from a specific IP address but allow outgoing traffic to establish connections. Which firewall type is best suited for this based on connection state tracking? (1 Marks)

- A. Packet Filtering Firewall
- B. Stateful Inspection Firewall
- C. Application-level Gateway
- D. Circuit-level Gateway

Q2. An organization needs to inspect SSL/TLS encrypted traffic and perform deep packet inspection to identify specific application-layer attacks. Which firewall type should they deploy? (1 Marks)

- A. Circuit-level Gateway
- B. Packet Filtering Firewall
- C. Next-Generation Firewall (NGFW)
- D. Stateful Inspection Firewall

Q3. A firewall operates at the Session layer of the OSI model, verifying the TCP handshake without inspecting the actual packet payloads. Which type of firewall is this? (1 Marks)

- A. Packet Filtering Firewall
- B. Circuit-level Gateway
- C. Application-level Gateway
- D. Stateful Inspection Firewall

Q4. A company wants to completely hide its internal network structure and force all external web traffic to terminate at an intermediary device that reconstructs the requests. Which firewall type meets this requirement? (1 Marks)

- A. Packet Filtering Firewall
- B. Circuit-level Gateway
- C. Application-level Gateway (Proxy)
- D. Stateful Inspection Firewall

Q5. A security engineer needs to protect a public-facing web server specifically from SQL injection and Cross-Site Scripting (XSS) attacks. Which firewall is most appropriate? (1 Marks)

- A. Network-level Firewall
- B. Web Application Firewall (WAF)
- C. Circuit-level Gateway

D. Packet Filtering Firewall

Q6. Why is a traditional packet filtering firewall considered vulnerable to IP spoofing attacks? (1 Marks)

- A. It does not inspect the payload of the packet.
- B. It only examines header information like source and destination IP without verifying authenticity.
- C. It operates at the Application layer.
- D. It maintains a state table of all active connections.

Q7. When a stateful inspection firewall receives an incoming packet, what is the first thing it checks to determine if the packet should be allowed? (1 Marks)

- A. The access control list (ACL) rules
- B. The state table of active connections
- C. The application-layer payload
- D. The digital signature of the packet

Q8. A remote worker's laptop needs protection against malicious traffic when connected to public Wi-Fi networks. Which type of firewall is most appropriate? (1 Marks)

- A. Network-based Firewall
- B. Host-based Firewall
- C. Circuit-level Gateway
- D. Cloud-based WAF

Q9. An enterprise is migrating its infrastructure to a multi-tenant public cloud environment and needs to filter traffic between virtual machines within the same virtual private cloud (VPC). Which firewall type is best suited? (1 Marks)

- A. Hardware Firewall
- B. Virtual (Cloud) Firewall
- C. Circuit-level Gateway
- D. Traditional Packet Filter

Q10. Which technology allows a firewall to analyze the actual data field of an IP packet to detect malware or policy violations? (1 Marks)

- A. Stateful Packet Filtering
- B. Deep Packet Inspection (DPI)
- C. Circuit-level Verification
- D. Simple Network Management Protocol

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Stateful inspection firewalls track the state of active connections to dynamically allow returning traffic.

Q2. Correct Answer: C

Rationale: NGFWs combine traditional firewall capabilities with deep packet inspection and application-level control.

Q3. Correct Answer: B

Rationale: Circuit-level gateways monitor TCP handshakes and session establishment at the Session layer without inspecting packet payloads.

Q4. Correct Answer: C

Rationale: Proxy firewalls act as intermediaries, terminating the connection and hiding internal network addresses.

Q5. Correct Answer: B

Rationale: WAFs specifically analyze HTTP/HTTPS traffic to protect web applications from exploits like SQL injection and XSS.

Q6. Correct Answer: B

Rationale: Packet filtering firewalls inspect only basic header information, making them susceptible to spoofed source IP addresses.

Q7. Correct Answer: B

Rationale: Stateful firewalls first check the state table to see if the packet belongs to an already established, authorized connection.

Q8. Correct Answer: B

Rationale: Host-based firewalls run locally on individual devices to protect them regardless of the network they connect to.

Q9. Correct Answer: B

Rationale: Virtual firewalls are deployed within cloud environments to monitor and secure east-west traffic between virtual machines.

Q10. Correct Answer: B

Rationale: Deep Packet Inspection looks beyond the packet headers to examine the actual payload data for security threats.